

Zubaidah Al-Mashhadani

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TECHNICAL SKILLS

Programming: Python, MATLAB, R, C, C++, Verilog, Assembly, Linux/Bash
ML & Signal Processing: PyTorch, TensorFlow, SciPy, NumPy, Pandas; dimensionality reduction, domain adaptation, Riemannian geometry, neuromorphic computing, quantization, distillation, computer vision, autonomous navigation, perception
Biosignal Acquisition: Surface EMG, EEG, Lab Streaming Layer (LSL), BrainFlow, MEEGkit
Hardware & Embedded: NVIDIA Jetson, Raspberry Pi, Arduino, OpenCR, FPGA, PCBs, BLE, Zigbee, Intel RealSense, LiDAR, piezoresistive tactile sensors
Robotics & Vision: ROS/ROS2, CUDA, Gazebo, OpenCV, SLAM, multimodal perception, sensor fusion, object detection/segmentation
Tools: Git, Docker, Jupyter, SolidWorks, AutoCAD, Simulink, MultiSim

EDUCATION

University of Central Florida Orlando, FL
Ph.D., Electrical Engineering — LIMB Lab (Advisors: Dr. Mohsen Rakhshan, Dr. Di Wu) Aug. 2021 – Aug. 2026
• Dissertation: *Closed-Loop Human-Machine Interfaces*
★ Ali Seyedi Reza Outstanding Dissertation Award

Florida Polytechnic University Lakeland, FL
M.S., Robotics Engineering | Fulbright Scholar (Advisor: Dr. Balasubramaniyan Chandrasekaran) Aug. 2019 – May 2021

Engineering Technical College Baghdad Baghdad, Iraq
B.S., Mechatronics Engineering (Advisor: Dr. Mazin Ismael) 2011 – 2015

EXPERIENCE

Postdoctoral Fellow Aug. 2026 – Present
LIMB & Unary Lab, University of Central Florida (Advisors: Dr. Mohsen Rakhshan, Dr. Di Wu) Orlando, FL
• Developing uSense, a unary-computing-based stochastic neuromorphic sensing framework for ultra-low-power on-device biosignal compression and inference (*Nature Communications AI & Computing*, under review).
• Developing novel spike detection framework leveraging unary computing for ultra-low-power on-device neural signal processing; targeting real-time, edge-deployable biosignal inference.

Graduate Research Assistant Aug. 2021 – Aug. 2026
Laboratory for Interaction of Machine and Brain (LIMB) & Unary, UCF Orlando, FL
• Developed robust surface-EMG decoder agnostic to limb position, recording session, and user one-shot latent representation alignment (IEEE TNSRE 2025); designed multimodal LSL data acquisition setup (JNE 2026).
• Designed closed-loop tactile feedback systems for robotic prosthetic fingertips using piezoresistive sensors; demonstrated robust performance across 21 surface textures with low-latency on edge.
• Built EEG-based emotion recognition system achieving 9% accuracy improvement via lower-dimensional Riemannian manifold feature extraction; improved feature selection by 5% (*Applied Sciences* 2023).
• Reviewed manuscripts for IEEE RA-L (5 reviews) and IEEE BSN (2 reviews); mentored 30+ undergraduate students.

Graduate Research & Teaching Assistant Aug. 2019 – May 2021
Florida Polytechnic University (Advisor: Dr. Balasubramaniyan Chandrasekaran) Lakeland, FL
• Developed computer vision pipeline for tomato disease and ripeness detection for an agricultural robot (IEEE IEMCON 2021).
• Developed a perception and SLAM pipeline using camera and LiDAR data for collision-free autonomous robot navigation and human-in-the-loop interaction. (IEEE ICCR & IEEE UEMCON 2020).

SELECTED PUBLICATIONS

Al-Mashhadani, Z., Rakhshan, M., Wu, D. (2026). uSense: Unary-Computing-based Stochastic Edge Neuromorphic Sensing. *Nature Communications AI & Computing* (under review).

Al-Mashhadani, Z., Overton, T., Wu, D., Rakhshan, M. (2025). Towards Plug & Play Myoelectric Control via One-Shot Latent Representation Alignment. *IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)*.

Overton, T., **Al-Mashhadani, Z.**, Raza, S.Y., Whitson, J., Rakhshan, M. (2025). Limb position effect in myoelectric control: Strategies for optimisation and standardisation. *Journal of Neural Engineering (JNE)*.

Al-Mashhadani, Z., Bayat, N., et al. (2023). Efficacy of lower-dimensional Riemannian geometry for EEG-based emotion classification. *Applied Sciences*, 13(14), 8274.

Additional publications: IEEE ICCAS 2023, Journal of Imaging 2023, IEEE IEMCON 2021, IEEE ICCR 2020, IEEE UEMCON 2020, IEEE IEMCON 2020.

REFERENCES

Mohsen Rakhshan, Assistant Professor, UCF | mohsen.rakhshan@ucf.edu | (407) 823-2044

Di Wu, Assistant Professor, UCF | di.wu@ucf.edu | (407) 823-1341